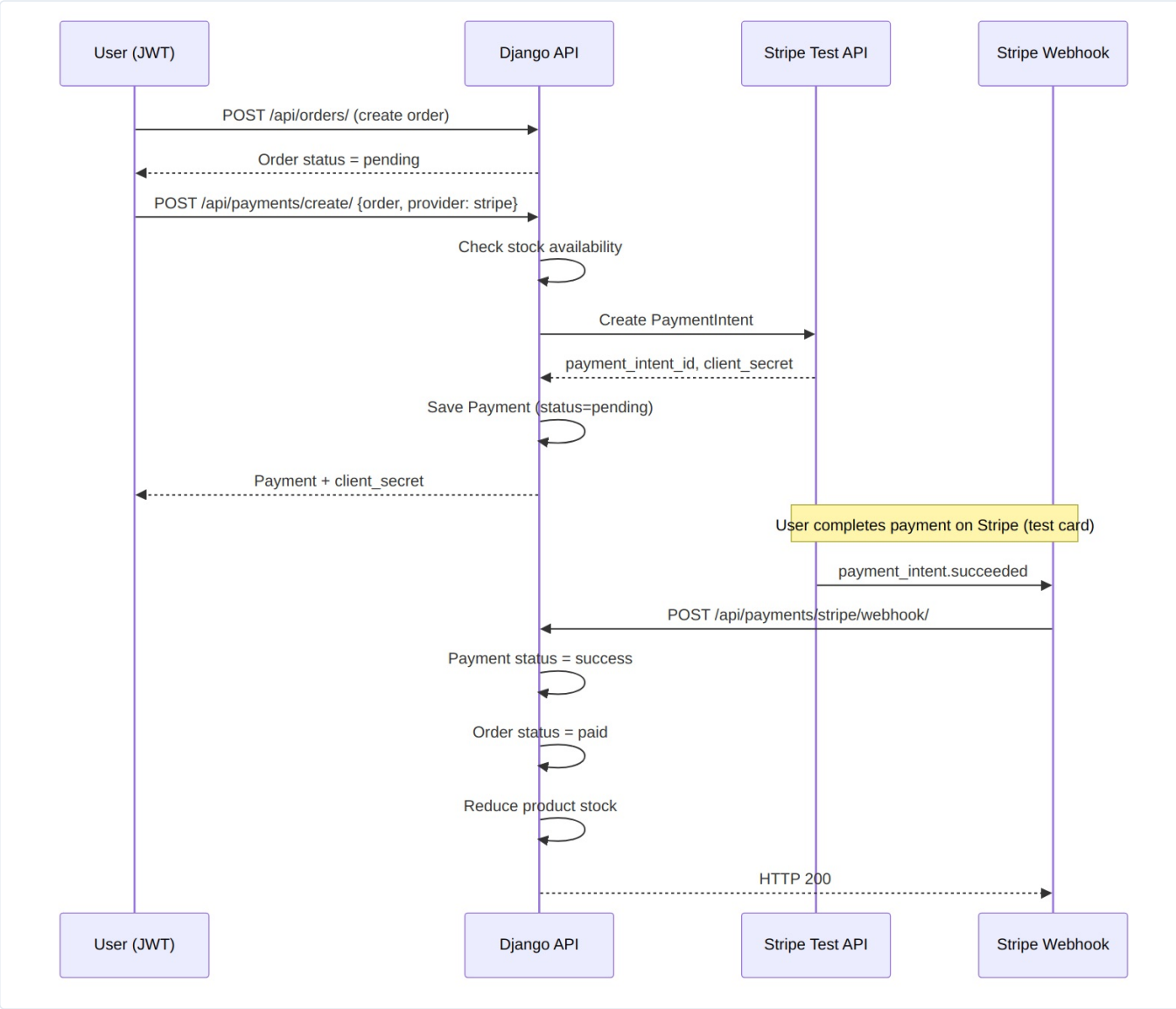


# Payment Flow Diagrams

# Payment Flow Diagrams

## Stripe Payment Flow (Test Mode)



### Stripe Test Setup

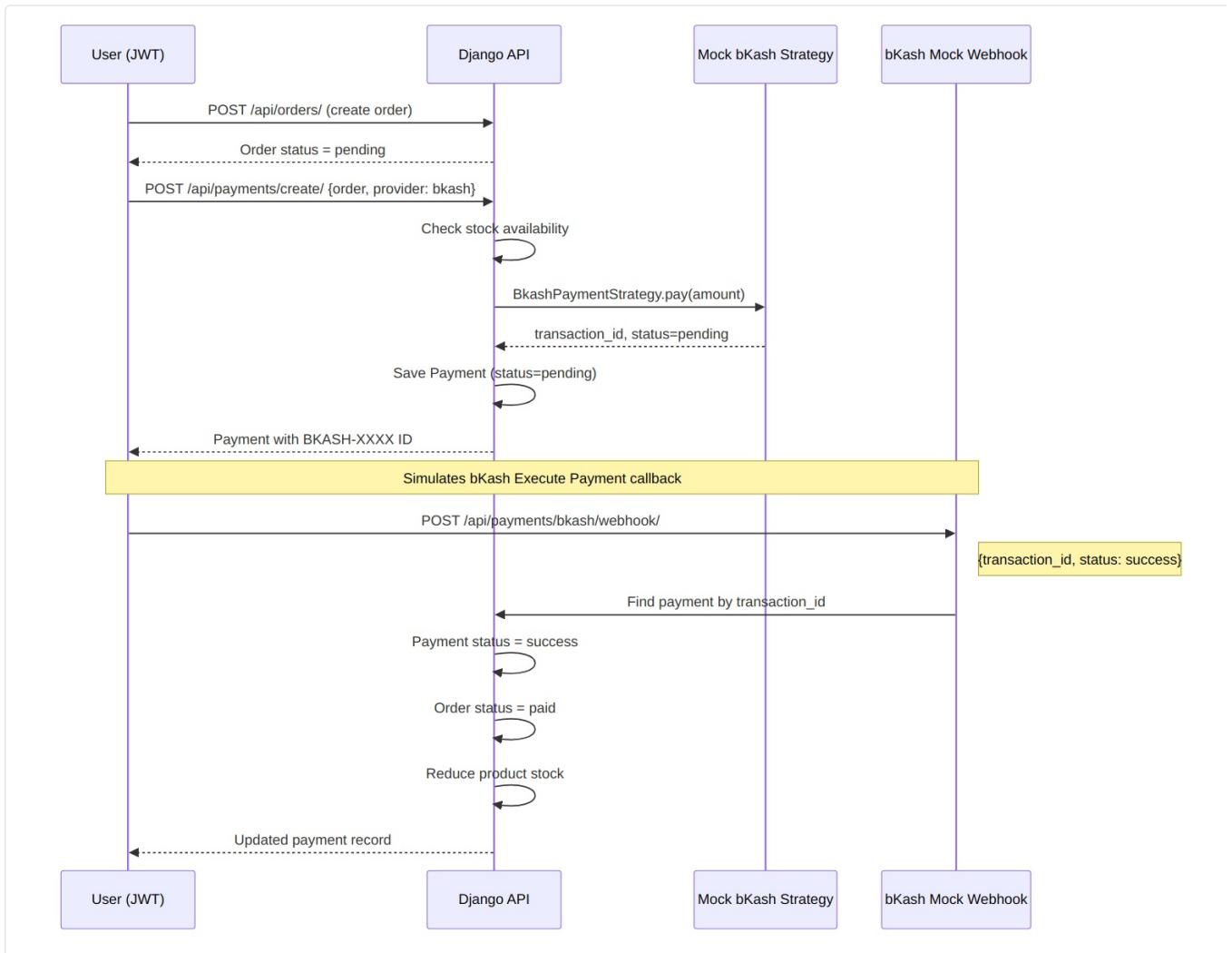
```
stripe login
stripe listen --forward-to http://127.0.0.1:8000/api/payments/stripe/webhook/
stripe trigger payment_intent.succeeded
```

### Test Card

- Number: 4242 4242 4242 4242
- Any future expiry, any CVC

## bKash Mock Payment Flow (Sandbox Simulation)

**Note:** Real bKash sandbox/live APIs require a bKash merchant account. This project implements a mock flow that mirrors the real checkout → execute → confirm pattern using the Strategy Pattern.



## Mock bKash Webhook Example

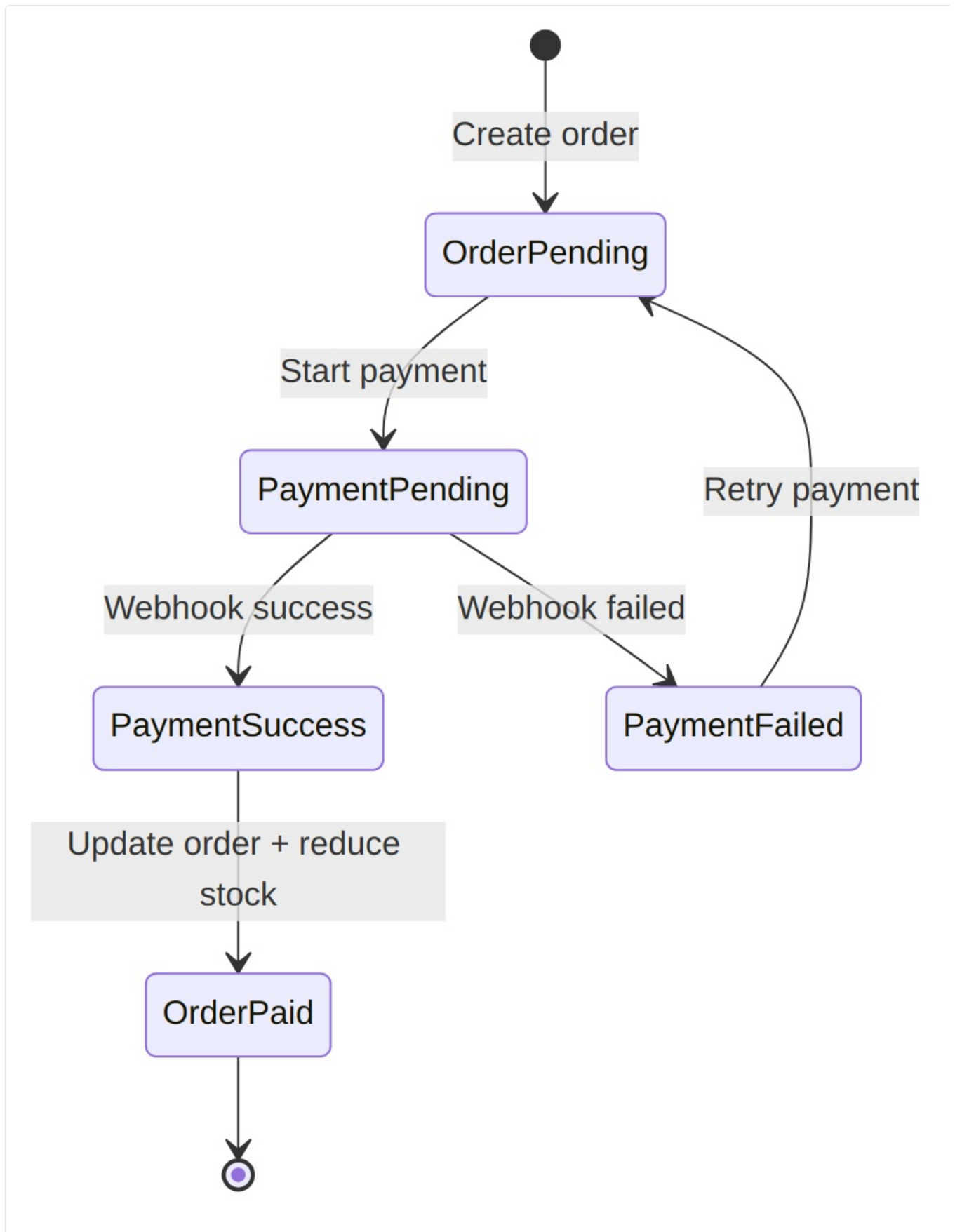
```
POST /api/payments/bkash/webhook/
Content-Type: application/json

{
  "transaction_id": "BKASH-XXXXXXXXXXXX",
  "status": "success"
}
```

To simulate failure:

```
{
  "transaction_id": "BKASH-XXXXXXXXXXXX",
  "status": "failed"
}
```

## Complete Order Lifecycle



### Stock Management Rules

| Stage | Action |
|-------|--------|
|       |        |

|                 |                                              |
|-----------------|----------------------------------------------|
| Order create    | Validate stock $\geq$ quantity for each item |
| Payment create  | Re-check stock before initiating payment     |
| Payment success | Atomically reduce stock (webhook handler)    |
| Payment failed  | Stock unchanged                              |

Stock is **never** reduced until payment status becomes `success` .